Boundary Deployment and Contact Feasibility in Cooperative Transport without Prior Object Geometry

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***Abstract*— How far does deployment from local boundary observations carry a robot team toward actual object transport? We study the perception–deployment–contact–transport–stop chain without an a priori complete object model. Three deployment modules share sensing, downstream control, unilateral contact physics and independent completion rules. A nine-condition census exposes orientation-dependent failures and a 1.177 m spatial miss despite projected arrival. In a later matched comparison of 12 conditions, boundary CVT improves common-interval coverage by 1.52 and 4.85 percentage points over local spacing and nearest observed targets, while all three complete 11/12 tasks. A finite-boundary constraint candidate fails all four trials whose controls complete. Verified optimization witnesses exclude forward velocity in 11 of 17 saved positive-force states, while other states admit forward motion but select reverse commands. These dependent states do not establish complete trajectory causation. A separate 60-episode validation on one new seed exposes two completion failures with eight robots and two penetration-budget violations with 24. The contribution is a reproducible characterization of the deployment-to-transport gap, with restricted mathematical explanations and preserved negative results. Dynamic safety, arbitrary-shape reliability and fully decentralized supervision remain unsupported.**

I. Introduction

A ring of robots around an unknown object is a preparation state, not a completed manipulation task. The team must perceive usable boundary information, distribute itself, generate actual contact forces, move along a requested displacement, and stop near a spatial goal. Failures can occur after apparently satisfactory coverage. Our research question is therefore: *under fixed sensing and transport control, how does boundary-based deployment affect executable transport across geometry and orientation?*

This question must be distinguished from novelty by demonstration. Cooperative transport and pushing span substantial prior work [18], [19]. Caging-based transport already addresses unknown shapes [2], and adaptive self-allocation has combined local detection, CVT and transport [1]. We do not claim that surrounding several shapes or using CVT is new. Instead, this development manuscript presents a controlled evaluation of an existing boundary-aware stack and identifies the evidence still needed for a defensible method contribution.

The deployment comparison changes only the target before the first CONTACT_READY event. Boundary CVT, nearest observed targets and local spacing otherwise share the same code and inputs. We connect a 27-task geometry census, a later 12-condition matched comparison, fixed contact-capability diagnostics, and a new 60-task resource/noise validation. Each batch retains its own source identity, stopping rules and complete denominator. The question is falsifiable: do deployment gains translate into more safe spatial completions under the same downstream control? These fixed-condition comparisons can fail; they are not population-level reliability tests.

Our contributions are an explicit separation of information and task contracts, a matched measurement of deployment benefits against complete-task outcomes, and a reproducible diagnosis of constraint-induced loss of directional capability. The optimization certificates are standard convex bounds applied to saved controller states, not a new safety theorem. We test and reject mechanisms rather than infer task success from geometric coverage or QP feasibility. An earlier progress oracle rescued two selected failures, but a realizable anchored candidate did not. A later finite-boundary candidate regressed successful transport. We retain both negative branches and their distinct evidence units. This is an empirical research draft; a superior new transport algorithm has not been established.

II. Related Work and Claim Boundaries

Caging restricts object escape through configuration-space geometry [6], [10]. Local-observation enclosing and grasping [3] and sequential caging [2] are particularly close to the present task. We use *boundary deployment* or operational enclosure for the measured arrangement. A surrounding image or high sampled coverage alone does not establish object closure.

Planar pushing [4], [9], adaptive collaborative control [5], [7], Force-ANTS [8], and grasp-wrench quality [20] connect coordination to manipulation under different assumptions. Sequential caging includes effective pushing and contact maintenance under initialization and communication assumptions. Song et al. combine detection-driven density, CVT and CBF coordination; their reported simulation setting differs from the present polygon/contact census. Tang et al. optimize contact modes and wrench feasibility using known object geometry and physical properties. These precedents preclude claiming novelty for local detection, CVT, or task-directed contact selection alone. Our two engineering deployment controls neither reproduce these approaches nor form a state-of-the-art benchmark.

CVT and coverage control provide the deployment background [14], [15]. Safety QPs relate to control-barrier methods [16], [17], but moving uncertain boundaries and sampled contact physics require assumptions beyond static-map results. Registration covariance and stable sampling [11], [12], [13] motivate the restricted information audit below. Neither

a well-conditioned local matrix nor a correct numerical identity proves physically accurate cumulative progress.

III. System, Information and Task Contracts

A. Local deployment with shared supervision

Each robot observes nearby boundary samples and forms a local fused map using its own sensing and one-hop observation exchange. The nominal deployment follows a boundary-weighted centroid in a local Voronoi region. In generic notation,

$$c_i = \frac{\int_{V_i} q\rho_i(q)\,dq}{\int_{V_i} \rho_i(q)\,dq}, \qquad u_i^{\text{deploy}} \propto c_i - p_i, \tag{1}$$

where the implemented density includes observed-boundary, directional-offset and exploration terms. The census preserves the existing density, discretization, gains and guards. The QP modifies the nominal command subject to the existing inter-robot and observed-object constraints; no thresholds are changed to improve outcomes.

The nearest control substitutes the closest observed offset target. The spacing control selects among locally eligible offset targets using neighbor separation and distance. Both preserve the CVT mass and deployment guards. After the first CONTACT_READY event, all three use the same subsequent controller. Earlier deployment can change contact configurations and downstream trajectories; shared code does not imply identical post-deployment states.

Local transport feedback uses the robot's own and neighboring usable progress estimates. A team-wide median progress and aggregate map coverage, readiness and speed signals drive a common monotone phase supervisor. The DBACT project name refers to decentralized boundary-aware cooperative transport as the research direction; this implementation does not yet establish a fully decentralized end-to-end loop. Shared task direction, neighbor communication and common phase state remain explicit assumptions. Object truth is available to simulated sensors, physics and independent evaluation, not to the local control readout. The earlier oracle is a separately labeled exception used only for diagnosis.

B. Unilateral contact and sampled execution

Robots are kinematic discs with commanded velocity bounded by $V = 0.35$ m/s. The rigid object's polygon is known to the physics engine and evaluator, but not provided to local transport control. Let s_i be the signed surface distance at robot center p_i, n_i the outward normal at its closest boundary point q_i, and $v_i^{\text{rel}} = u_i - v_o(q_i)$. For penetration $\delta_i = r - s_i > 0$, the implemented penalty model uses

$$f_{n,i} = [k_n\delta_i - c_n n_i^T v_i^{\text{rel}}]_+, \tag{2}$$

$$f_i = -f_{n,i}n_i + \min(\mu f_{n,i}, k_t\|v_{t,i}\|)\frac{v_{t,i}}{\|v_{t,i}\|}, \tag{3}$$

where $v_{t,i} = v_i^{\text{rel}} - (n_i^T v_i^{\text{rel}})n_i$ and the tangential term is zero for negligible slip. There is no attractive attachment force. The physical parameters are $r = 0.13$ m, $k_n = 500$ N/m, $c_n = 12$ N s/m, $\mu = 0.6$ and $k_t = 60$ N s/m. Torque is $(q_i - x) \times f_i$. Four substeps per 0.05 s control interval integrate the summed forces and torques; Coulomb ground-friction impulses are applied by velocity clamping, followed by the configured speed limits. Robot endpoints are updated first and held during these cargo substeps. Thus saved contact force is neither a pre-command measurement nor a continuous contact-safety proof.

The common transport loop forms a declining speed reference $v_{\text{ref}} = \text{clip}(k_b(L - \widehat{J}), 0, v_0)$ and PI effort $a = \text{clip}(k_p e + k_i z, 0, a_{\max})$, where $e = v_{\text{ref}} - d^T\widehat{v}_o$. Integration is conditional on blocking and a deadband, with bounded integral and saturation back-calculation. Braking and HOLD can reset the integral or suppress effort. The nominal command combines deployment/station-keeping, estimated convoy motion and selected inward pressing. Later batches add observed-reference vector navigation and contact allocation; they are not interchangeable with the earlier scalar-feedback census. All executable configurations and disabled experimental switches are versioned with each batch.

C. Five distinct events

CONTACT_READY uses a local-map distance criterion and a team quorum/dwell. In the frozen configuration, the distance band extends to 0.185 m whereas robot radius is 0.13 m. It therefore permits a geometric-ready state without physical contact. Physical contact requires penetration and contact force in the simulator. A separate recorded physical-readiness event requires sustained contacts with positive force projected on the task direction. It is stronger than the occurrence of one actual contact and is not assumed necessary for completion.

For target direction d, displacement L, initial object centroid x_0, and terminal centroid x_f,

$$J(t) = d^T(x(t) - x_0), \qquad e_{\text{end}} = \|x_f - (x_0 + Ld)\|\,. \tag{4}$$

Settling checks linear speed only, with no angular-speed criterion. It requires HOLD and speed at most 0.005 m/s for 40 consecutive control frames. Spatial completion requires settling *and* $e_{\text{end}} \le 0.12$ m. The control step is 0.05 s. The historical census has a 3000-frame watchdog; the later batches below freeze a 12000-frame budget before execution. A program finishing successfully is not a task completion.

G500 is a legacy execution contract, not a stricter version of spatial completion. Here it requires specified events and execution/safety checks, but arrival uses only $|J - L| \le 0.12$ m. Efficiency, direction error, cross-track deviation and yaw are diagnostic quality warnings, not pass/fail gates. Event deadlines are disabled, so its name does not impose a 500-frame limit. We report completion, G500 and trajectory quality separately. Discrete safety measurements and barrier-scaling counts are not continuous-time safety certificates.

IV. Frozen Full-Task Development Census

A. Geometry and fair comparison

The nine conditions are circle; ellipse long axis parallel/perpendicular to the task; rectangle long side paral-

lel/perpendicular; L reference/rotated 180°; and C opening toward/away from the goal. All world task directions are $+x$. The current CCTA rectangle, L and C outlines are retained as sources. The current C-channel opens in $+y$; rotations -90° and $+90^\circ$ produce the two conditions. L reference means its unrotated source polygon, not an ambiguous opening label. The frozen vertices and figure record each geometry.

Circle and ellipse reuse the historical deterministic 64- and 24-vertex outlines, rather than importing their old scenario settings. Every polygon is centroid-centered at $(4, 4)$ and isotropically normalized to area 2.05875 m^2. Areal density is 2 kg/m^2, giving mass 4.1175 kg. Actual polygon inertias are approximately 1.349, 1.550, 1.472, 2.087 and 2.547 kg m^2 for circle, ellipse, rectangle, L and C. Perimeter, reach, inertia and concavity remain different; equal area and mass do not equalize manipulation difficulty.

All conditions use 16 robots and the same legal scatter initial positions, generated once by seed 2 in a common annulus outside the maximum normalized reach. Actual initial-state and configuration hashes are verified for each three-method group. The goal distance is 1.2 m. Sensing, robot size, actuation, friction, contact stiffness, four physics substeps, control parameters and success rules are common. The team starts near the object, with some robots initially in sensing range: unknown full geometry is tested, not search from an entirely unknown remote location.

B. Budget, records and selection

The budget is exactly $9 \times 3 \times 1 = 27$ episodes. We do not replace seeds, retry failed tasks, extend watchdogs or tune parameters against outcomes. Source/environment snapshots, resolved configurations and product hashes identify this census separately from historical batches and v1/v2.

Every case records phases, contact, motion, braking, termination and safety. All 27 started tasks remain in the table. The nine conditions are within-block descriptive comparisons. The independent randomization unit is the complete shared scene/seed block, of which this census contains one. Frames and contact records are not independent trials; no reliability interval is reported.

Before execution, we select boundary CVT under the five reference orientations for the stage figure and overview video. All nine conditions have three-method videos synchronized by physical time and common view. Ended episodes retain their final measured frame with an explicit label. Videos use every fifth control frame at 10 fps (nominal $2.5\times$ playback), not episode-fraction synchronization. Estimated progress is labeled as the previous control update relative to the post-physics displayed pose; the new census mechanism figure instead aligns the estimate with the preceding, pre-physics object state. Raw laser rays are not retained: displayed teal samples are agent 00's saved fused map.

V. Capabilities and Failure Boundaries

Table I reports every condition and method; the supplement adds actual termination time and total robot path. Boundary CVT completes 5/9 tasks, nearest target 2/9, and local spacing 5/9. Corresponding G500 counts are 5/9, 1/9 and 3/9. These fixed-condition counts characterize this implementation; they are not success-rate estimates for arbitrary objects.

TABLE I

All nine conditions. Entries: endpoint error in meters / spatial completion (T or F). One fixed seed per condition; three methods share initial states. Full records additionally contain stage times, safety and G500 reasons.

Condition	CVT	Nearest	Spacing
Circle	1.177 / F	1.214 / F	0.906 / F
Ellipse parallel	0.838 / F	1.328 / F	0.903 / F
Ellipse perp.	0.356 / F	0.321 / F	0.394 / F
Rect. parallel	0.096 / T	0.217 / F	0.061 / T
Rect. perp.	0.039 / T	0.097 / T	0.030 / T
L reference	0.112 / T	0.134 / F	0.117 / T
L rotated	0.120 / T	0.306 / F	0.112 / T
C toward	0.101 / T	0.049 / T	0.046 / T
C away	0.846 / F	0.496 / F	0.818 / F

The proposed descriptive superiority criterion is not met: CVT ties local spacing in completion count. Circle and both ellipse orientations fail spatial completion under all three methods. Both rectangle orientations complete under CVT and spacing, while the nearest-target control completes only the perpendicular rectangle. L reference and rotated L complete under CVT and spacing; nearest-target fails in both. All methods complete with the C opening toward the goal, whereas all three C-away tasks exhaust the original 150 s watchdog and fail. The C-away endpoint errors are 0.846, 0.496 and 0.818 m for CVT, nearest and spacing. These results show a task-orientation boundary under this common configuration, not a claim that C shapes are intrinsically impossible.

In total, 22 episodes settle and five reach the watchdog: ellipse-parallel/spacing, rotated-L/nearest and all three C-away cases. Every episode reaches geometric CONTACT_READY and records physical contact and positive projected motion; none of those events suffices for completion. CVT and spacing achieve completion in the same five conditions but with different endpoint errors. These data support examining lateral motion and termination alongside deployment, rather than selecting a new feedback candidate solely from rank deficiency.

The circle/CVT example makes the contract distinction concrete. It reaches geometric readiness, produces actual contact and motion, and settles after 13.50 s. Its projected progress is 1.1247 m, within 0.12 m of the 1.2 m target. Its remaining hard execution gates also pass, so G500 passes. Yet lateral displacement yields $e_{\text{end}} = 1.1772$ m. Recorded direction error is 46.25°, efficiency 0.6915, and maximum cross-track 1.1748 m. These create quality warnings, not G500 failures. Changing the contract to hide this discrepancy would destroy comparability; we retain both labels.

The supplementary five-shape event plate retains the preselected reference orientations and failures; all nine orientations have synchronized three-method videos. First physical

contact is selected independently of the supervisor events. The figure sequence is event-indexed rather than an assumed chronological pipeline.

Figure 1 separates the actual team-progress readout, signed estimation error and true cross-track displacement. Census replay row $k+1$ stores the estimate computed before that step's physics; we therefore compare it with object truth at row k. All control samples are retained without smoothing. The circle approaches its projected goal while developing a large lateral offset; the C-away case instead stalls with a large longitudinal shortfall. These distinct trajectories do not establish a common causal estimation fault. Actual contact reports are from after robot motion but before cargo integration, and are labeled separately from control-time states.

VI. Matched Evidence and Current Validation

A. Deployment coverage without a completion advantage

A subsequent frozen comparison evaluates three deployment modules and two already selected motion-estimation speed priors, 0.25 and 0.45 m/s, across 12 new-start conditions. Eight are near-field and four begin with no scan returns or initial-location hint; the shapes also include a U-tray and a skew box. One seed, 20261102, is shared across the designed conditions. Each six-arm block has identical robot positions and task. The prior is an estimator parameter, not the safety object's speed bound. These 72 trials are distinct from the census and from the 33-condition historical regression; they cannot be pooled as repetitions of one controller.

For CVT, the prior change yields spatial and hard-safety completion 11/12 to 11/12 and G500 9/12 to 9/12. It therefore fails the predeclared additional-completion gate, despite reducing mean endpoint error. Under the common 0.45 setting, all three deployment modules complete 11/12; G500 is 9/12, 6/12 and 8/12 for CVT, spacing and nearest, respectively. No hard safety failure is recorded in those 36 episodes. These figures do not establish equivalence or reliability over a distribution of unseen objects.

A post hoc audit of all 36 trajectories measures coverage during a common prefix: each block ends at the earliest CONTACT_READY time among its three methods. Twenty equal time intervals and an independent 360-point boundary discretization give time-averaged coverage and maximum uncovered-arc diagnostics. This avoids comparing coverage at three different readiness times, although the common horizon remains outcome-dependent. Relative to spacing and nearest, CVT increases mean coverage by 1.52 and 4.85 percentage points and reduces mean maximum gap by 0.052 and 0.190 m. Its mean per-robot path changes by -0.129 and -0.013 m. Readiness is 1.01 s later than spacing and 0.33 s earlier than nearest. All blocks and tradeoffs are retained in Fig. 2; no frame-level significance test is used.

B. A finite-boundary candidate regresses transport

An independently frozen eight-episode development comparison then replaces extrapolated aggregate object rows with finite current-scan support after reference acquisition. It retains the existing numerical parameters, one-hop packet permissions, physical model and completion rules. The four conditions are near circle, near L, far C-away and far skew box. These conditions have been used repeatedly for development; the experiment is prospective with respect to this candidate, not an unused generalization set. All four map-bridge controls complete and reproduce their archived scientific records. All four candidates reach the 600 s budget without completion, with endpoint errors 1.041, 1.142, 1.197 and 1.195 m. G500 changes from 2/4 to 0/4. No original hard-physical-safety gate is triggered, but this is not evidence that the new unrelaxed boundary inequalities hold throughout execution.

The complete per-frame constraint replay was interrupted and remains incomplete. We therefore make no full-batch constraint acceptance claim. Instead, a separate fixed-state audit selects the first and peak positive projected contact-force events after reference freezing and before braking in each control: eight events, 128 robot states, 112 admitted by the candidate, and 17 with positive projected force. Exact replay of 1749 control frames recovers these inputs without integrating a new physical trajectory. The force labels belong to the original command's subsequent contact evaluation; they are not assigned to counterfactual velocities.

C. Feasible direction and selected direction are different

For a saved robot state, let the complete linear rows be $Au \geq b$, with the unchanged speed ball. Define

$$\mathcal{F} = \{u : Au \geq b,\ \|u\| \leq V\}, \tag{5}$$

$$u^* = \arg\min_{u \in \mathcal{F}} \|u - u_{\mathrm{nom}}\|^2, \qquad \alpha = \max_{u \in \mathcal{F}} d^T u. \tag{6}$$

The first optimization selects the least-deviation command; the second asks whether any forward command exists. Feasibility alone does not answer the latter. For any $\lambda \geq 0$ and any feasible u,

$$d^T u \leq V\|d + A^T\lambda\| - b^T\lambda. \tag{7}$$

Indeed, adding $\lambda^T(Au - b) \geq 0$ and applying Cauchy–Schwarz proves the bound. A negative upper bound excludes every forward velocity. A multiplier with $V\|A^T\lambda\| - b^T\lambda < 0$ certifies infeasibility. These are ordinary convex separation witnesses, not guarantees about object force or transport.

Even $\alpha > 0$ and $d^T u_{\mathrm{nom}} > 0$ do not ensure $d^T u^* > 0$. For example, with $d = (1, 0)$, $u_{\mathrm{nom}} = (0.1, 0)$ m/s, $-u_x + u_y \geq 0.2$ m/s and $V = 0.35$ m/s, the projection is $(-0.05, 0.15)$ m/s although $(0.01, 0.21)$ m/s is feasible and forward. This illustrative algebraic example is not a simulated episode. If instead a verified upper bound is negative, changing only the objective cannot restore forward capability.

Across all 336 admitted state/system combinations, planar enumeration and an independent conic solver agree, with maximum primal–dual gap 1.66×10^{-8} m/s. In the 17 positive-force states, the original aggregate system permits forward motion in all 17. The finite capped system permits it in six, forces reverse motion in ten, and is infeasible in one (Fig. 3). These 11 excluded states cannot be repaired solely by changing the objective. Three other saved states admit

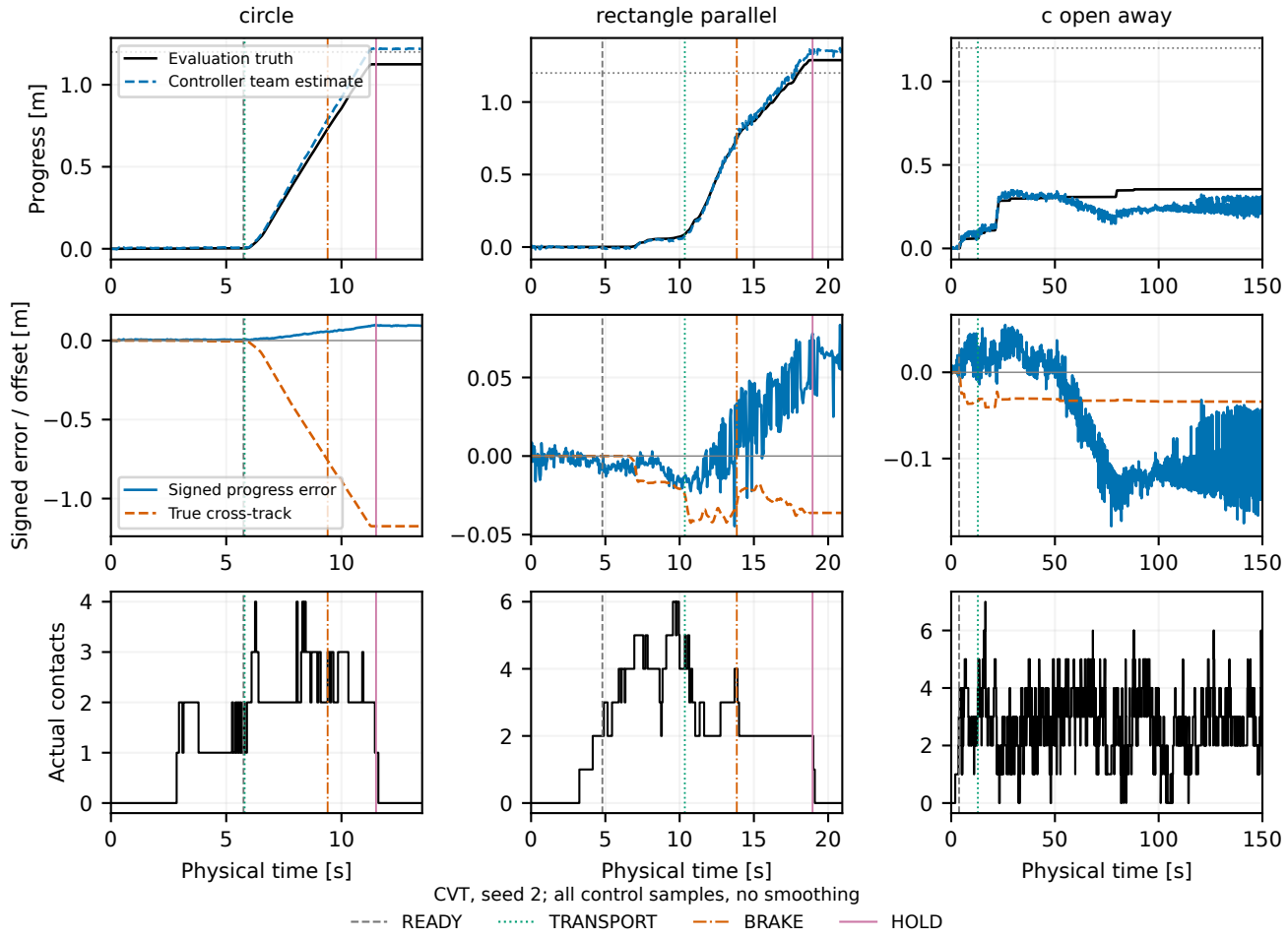


Fig. 1. Time-aligned CVT diagnostics for the circle, rectangle-parallel control and adverse C-away condition, selected after the census to illustrate distinct outcomes. Top: pre-physics evaluation truth and the actual shared team-progress estimate. Middle: signed progress error and true cross-track displacement. Bottom: actual contact reports. Event lines identify READY, TRANSPORT, BRAKE and HOLD; absent events are not imputed. Each column retains its actual time horizon, including the 150 s failure. One seed, no smoothing and no independent-sample inference.

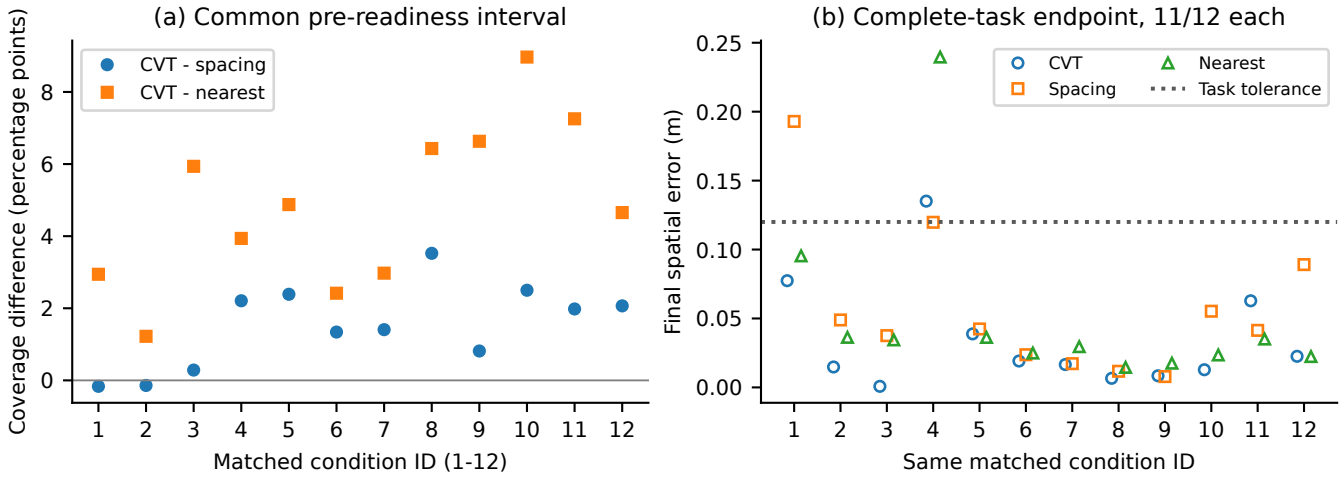


Fig. 2. Post hoc audit of all 36 matched trajectories under the common 0.45 m/s estimator prior. Left: CVT minus each deployment control over the same pre-readiness prefix within each condition. Right: final spatial errors from complete episodes. Each method completes 11/12; geometric improvement does not establish completion superiority. Condition IDs and receipt hashes are supplied in the evidence index. One shared seed; no confidence intervals.

TABLE II

CURRENT-SOURCE SENSITIVITY. EACH CELL LISTS COMPLETION WITHOUT A HARD SAFETY FAILURE, THEN G500; BOTH OUT OF FIVE. ONE SHARED SEED; DESCRIPTIVE COUNTS.

Regime	CVT	Spacing	Nearest
Reference	5/5; 5/5	5/5; 5/5	5/5; 4/5
8 robots	5/5; 3/5	4/5; 2/5	4/5; 3/5
24 robots	4/5; 1/5	4/5; 3/5	5/5; 1/5
Noise $\sigma = 0.02$ m	5/5; 5/5	5/5; 2/5	5/5; 3/5

forward velocity but the least-deviation projection selects reverse, including two declared pushers. States and alternative systems are dependent diagnostic units, not independent trials or a complete causal attribution for the four failed trajectories.

The finite rows subtract structural and three-sigma range allowances from geometric distance, and use received motion magnitudes in their demand. Neither three-sigma Gaussian noise nor received motion is a deterministic unknown-world bound. Ray-direction-aware uncertainty improves some numerical values but changes none of the fixed forward/reverse/infeasible classifications; that candidate also fails its sufficient-repair criterion. A set-valued local polygon admission design would require explicit geometric priors, a complete occupancy enclosure and a swept motion bound. Those components are not implemented and are not claimed as a contribution of the executed controller.

D. Current-source resource and noise validation

The current-source protocol records all 60 planned episodes on one new shared seed, with no outcome-driven retries. The five conditions are circle, ellipse-perpendicular, rectangle-parallel, L-reference and C-away. Each regime has five tasks per deployment module. Eight- and 16-robot layouts are prefixes of the same 24-robot layout; the noisier regime shares the reference initial positions. Fixed gains, four-pusher quorum and 600 s limits are retained. The full case table retains execution errors, watchdogs, endpoint errors, contact and phase events, trajectory quality and safety measurements. Robot-resource changes test the fixed controller, including its unchanged quorum; they are not separately optimized team policies. These designed conditions support sensitivity statements, not a population robustness probability.

Table II and Fig. 4 report the complete batch. All 60 cases reach geometric readiness and actual contact; 58 spatially complete and 56 also avoid the original hard safety failures. With eight robots, C-away/spacing ends in BRAKE at 600 s despite a 0.110 m endpoint error: spatial proximity alone does not meet the frozen settling contract. C-away/nearest enters HOLD but stops 0.824 m from the goal. CVT completes that condition only at 571.55 s. With 24 robots, L/CVT and ellipse/spacing spatially complete but reach penetrations of 0.1204 and 0.1157 m, exceeding their original discrete budgets of 0.0975 and 0.1002 m. More robots therefore do not monotonically improve safety under the fixed controller. Every method completes all five doubled-noise tasks without a hard safety failure, although G500 differs. This limited perturbation does not establish robustness to arbitrary noise or initial conditions.

VII. CLOSED FEEDBACK DIAGNOSTIC BRANCH

A. Oracle and negative local candidate

Nine archived rectangle/L/C development cases all reached geometric readiness and settled, but only five completed. None met the recorded positive-force quorum/dwell, despite completion in five cases. This rejects that particular recorded event as an unexamined necessary gate; it does not deny the need for physical interaction.

A six-episode diagnostic selected rectangle seed 2 and L seed 8 as failures and C seed 8 as a successful control. Replacing only local/team scalar progress with truth, from episode start, reduced endpoint errors from 0.1621 to 0.0720 m and 0.1516 to 0.0485 m in the failures. C error increased from 0.0083 to 0.0135 m. The oracle is not deployable, and post-hoc selection limits interpretation. Matching readiness times does not imply identical states at readiness; the intervention acts over the whole trajectory.

The subsequent v2 audit captured actual trimmed registration normals and fusion corrections. Of 22,812 registrations, 22,797 were task-direction identifiable at the frozen numerical tolerance. All 15 remaining records were already rejected by the original quality gate. Thus strict rank deficiency did not explain admitted measurements in these cases. Identifiability still does not imply accurate correspondences, negligible rotation or low accumulated bias.

A separate fixed nine-episode v2 comparison tested original feedback, a passive certificate and a local anchored readout. Original and certificate-only were identical in all 25 replay arrays and full step fields. The anchored readout used no object truth and retained the existing gain, map, velocity, downstream median and rejection rules. It rescued neither failure. Rectangle endpoint error changed to 0.1592 m, L to 0.1522 m, and the successful C case worsened to 0.0187 m. Same-input local MAE was slightly worse in all three diagnostics. All nine G500 verdicts were false, including the spatially completed C cases. The candidate remains an experimental option, but expansion is stopped.

The restricted identifiability proposition and signed cumulative error accounting are retained in the supplementary source. They separate map increments, gain, held values and voter changes, but do not identify a unique physical cause or prove estimator accuracy.

VIII. LIMITATIONS AND RESEARCH DECISION

Each batch uses a shared robot arrangement and common $+x$ goals. The new validation seed was unused by the frozen protocols, but five familiar shapes and four designed regimes do not represent an arbitrary-object distribution. Equal area and mass leave inertia, perimeter and contact opportunities unequal. Penalty contact, a limited noise range, shared supervision and simulated localization restrict external validity. No hardware validation is claimed.

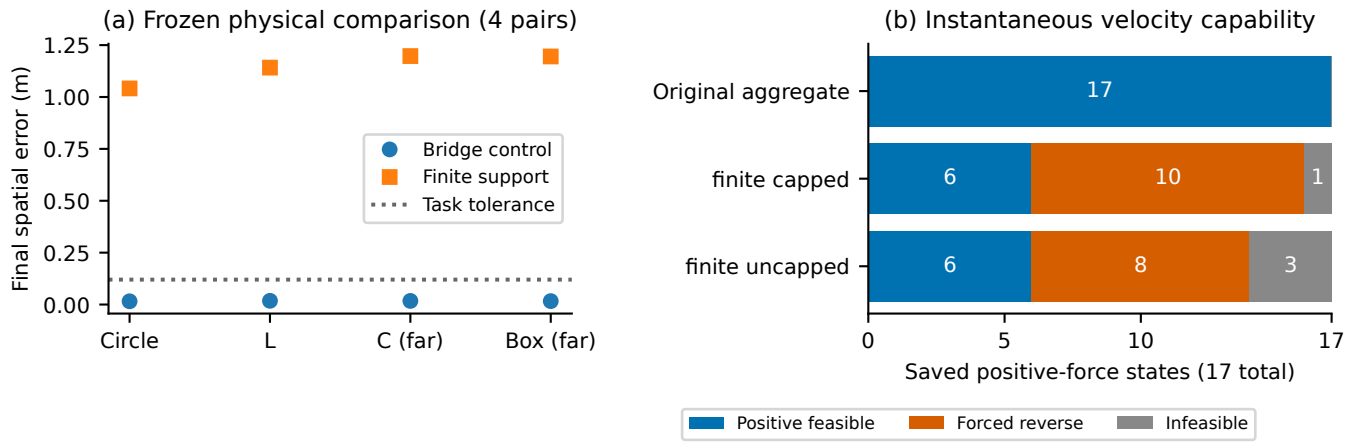


Fig. 3. Negative finite-support evidence with separate units. Left: every fixed physical pair; controls complete 4/4 and candidates time out 4/4. Right: instantaneous velocity capability in the 17 original positive-force states, under three complete constraint systems. Counts refer to the same states repeatedly. Capped means the implementation's reachable-demand truncation; it cannot be interpreted as satisfying the stronger untruncated rows. No counterfactual contact force or trajectory is fabricated.

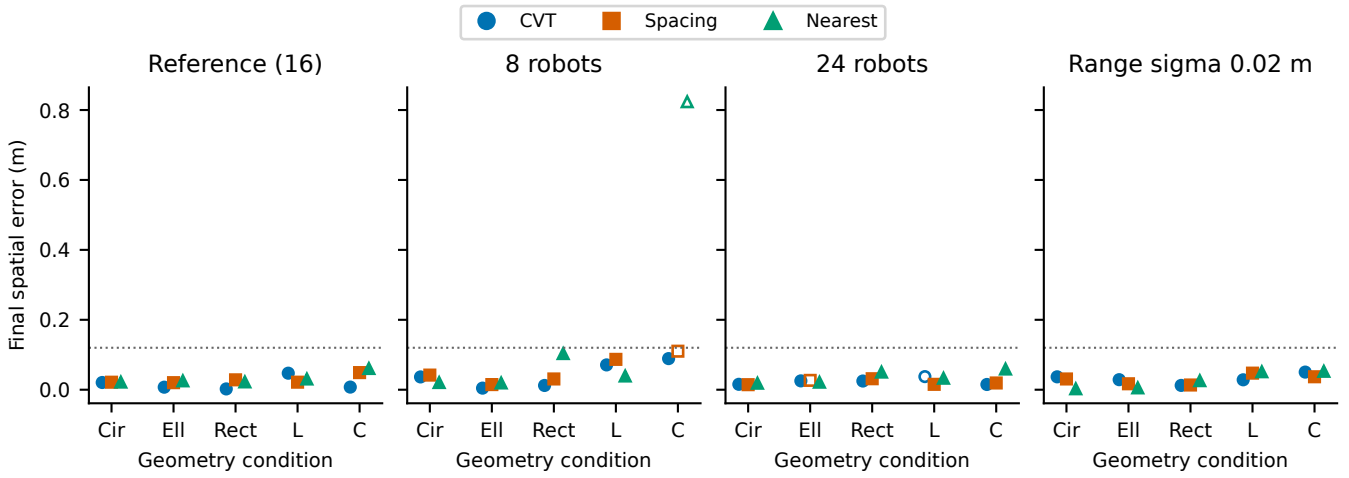


Fig. 4. All 60 current-source validation tasks, seed 2026110401. Each panel has five fixed geometries and three deployment modules. Filled markers denote spatial completion without an original hard safety failure; open markers include both incomplete tasks and safety violations. Thus an open marker below the 0.12 m line can represent a settling failure or excessive penetration, not missing data. Reference is 16 robots with range noise $\sigma = 0.01$ m. No outcomes are excluded or used to retune the controller.

Changed deployment also changes trajectories and contacts; it cannot assign all failures to feedback. Oracle/v2 cases cannot be pooled with this differently normalized census. Algebraic information and bookkeeping results establish neither transport stability nor continuous safety.

The progress branch is a mechanism diagnostic and limitation, not a new estimator contribution. The completed new-start comparison and the separate current-source sensitivity protocol do not validate arbitrary-shape robustness. Earlier source versions also have a 169-episode study, including 54 noise/communication/sensing perturbation trials: CVT, nearest and spacing complete only 1/18, 2/18 and 1/18 in that historical robustness population. Those failures and source snapshots remain available, but are not attributed to the current configuration. Further claims require matched information/action budgets and additional independent evidence. The anchored and finite-support candidates remain closed; their poor outcomes are not a reason to relax success criteria.

IX. Conclusion

Boundary deployment, contact-compatible motion and reliable stopping are separate requirements. The census exposes spatial failures hidden by projected arrival; matched later trajectories show coverage gains without a completion-count advantage. A finite-boundary constraint candidate regresses transport, and saved-state optimization witnesses identify a concrete directional obstruction while separating it from objective selection. These results support explicit task and information contracts, complete failure reporting, and checking both the permitted velocity set and its selected command. They establish neither a superior transport algorithm nor a dynamic safety theorem.

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